

EE491–EE492

Senior Design Project Midterm Report

# **Multi-Sensor Cough Analysis and State Classification with Machine Learning**

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10.11.2025

# 1 Introduction

Respiratory illnesses such as asthma, pneumonia, chronic obstructive pulmonary disease (COPD), and COVID-19 affect millions of people worldwide. Early detection and continuous monitoring are very important for accurate diagnosis and treatment. Traditional tools like stethoscopes have limitations as they depend on the doctor’s listening skills and the patient’s own reports. With advances in biomedical signal processing and machine learning, it has become possible to extract valuable information from physiological signals and cough sounds [1], [2].

This project will use a lab-built device that integrates four sensors: an accelerometer, a strain sensor, an ambient microphone, and a pulmonary microphone. Similar sensor-based systems have already shown promising results in cough detection and monitoring [3]. The goal of this project is to analyze cough events in detail by determining the patient’s physical state (sitting up, walking, running or standing up) during coughing and to detect whether the cough originates from the patient or from the surrounding environment.

## 2 Objectives

The main objectives of this project are:

1. To create a machine learning model that can distinguish between patient coughs and background coughs.
2. To identify the patient’s physical state during cough events using signals from an accelerometer, strain sensor, ambient microphone, and pulmonary microphone [3].

After getting satisfactory results from the model, I will compare and explore both classical machine learning and deep learning approaches [1], [4], compare single-sensor and multi-sensor performance, and if time permits, develop a basic real-time monitoring application.

## 3 Approach and Methodology

The project will begin with data collection and labeling using the four-sensor device. Recordings will be made while subjects perform activities such as sitting, walking, running, and resting. Cough events will be annotated manually and supported with automatic detection methods.

In the preprocessing stage, audio signals will be filtered and denoised, while accelerometer and strain data will be smoothed and normalized. Cough segments will be identified using short-time energy and spectral features [2]. All signals will then be divided into time windows to prepare for feature extraction.

For feature extraction, audio features such as mel-frequency cepstral coefficients (MFCCs), spectrograms, and spectral entropy will be used. From motion and strain data, statistical features and respiratory cycle patterns will be calculated [3].

For classification, several models will be trained, including Support Vector Machines (SVM), Random Forest, Gradient Boosting, and feedforward neural networks. Deep learning models like Convolutional Neural Networks (CNNs) or Long Short-Term Memory (LSTM) networks may also be tested, especially for spectrogram-based audio analysis [1], [4]. Sensor fusion methods will be applied to combine audio and motion data to improve overall accuracy.

Finally, the system will be evaluated using metrics such as accuracy, F1-score, sensitivity and specificity. Comparisons will be made between single-sensor and multi-sensor setups. To measure generalization, Cross-validation strategies such as k-fold validation and leave-one-subject-out testing will be used.

## 4 Work Completed

In accordance with the project proposal, the initial phase of the project has been completed. This phase included a comprehensive literature review on multi-sensor cough analysis and familiarization with the four-sensor recording device. Preliminary data acquisition sessions were conducted to gather an initial dataset.

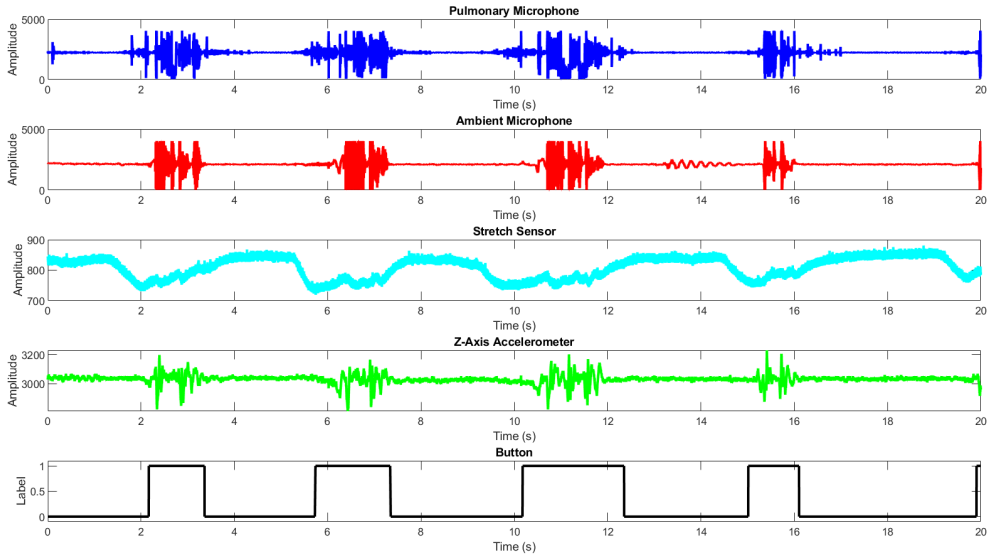


Figure 1: Sample visualization of the four-sensor data.

To analyze this data, visualization scripts were developed, with a sample output shown in Fig. 1. This visualization confirms the viability of the multi-sensor approach. As seen in the figure, cough events (e.g., around 2–3s, 6–7s) produce clear, high-amplitude bursts

in both the Pulmonary and Ambient Microphones. Crucially, these events correspond to sharp, high-variance signals in the Z-Axis Accelerometer and Stretch Sensor, validating their use for distinguishing patient coughs from ambient noise.

An analysis of the preliminary dataset was also performed, as summarized in Table 1. This analysis revealed a significant dataset imbalance. While 'Sittingup' (40 samples) and 'Walking' (20 samples) have a reasonable number of recordings, 'Running' (1 sample) and 'Standingup' (3 samples) are severely under-represented. This shows that a focused data collection effort is the most critical next step before proceeding to model training.

Table 1: Distribution of the initial collected dataset.

| activity            | , clean, | coughnoise, | musicnoise, | sneezenoise, | snoozenoise, | doornoise, | Total |
|---------------------|----------|-------------|-------------|--------------|--------------|------------|-------|
| sittingup           | , 18,    | 14,         | 4,          | 2,           | 1,           | 1,         | 40    |
| walking             | , 5,     | 13,         | 2,          | 0,           | 0,           | 0,         | 20    |
| running             | , 0,     | 0,          | 1,          | 0,           | 0,           | 0,         | 1     |
| standingup/standup, | 3,       | 0,          | 0,          | 0,           | 0,           | 0,         | 3     |

## 5 Work To Be Completed

The remaining work will follow the methodology outlined in the proposal, with the immediate priority being the expansion of the dataset to address the class imbalances identified in Table 1.

Following the completion of data collection, the preprocessing stage will begin. As planned, audio signals will be filtered and denoised, while accelerometer and strain data will be smoothed and normalized. Once preprocessed, the signals will be segmented using time windows to prepare for feature extraction. A comprehensive feature vector will be created, including audio features like MFCCs and spectral entropy, as well as statistical features and respiratory patterns from the motion and strain sensors.

With the feature set prepared, baseline machine learning models such as SVM and Random Forest will be trained and evaluated using k-fold cross-validation. This will be followed by the development and testing of deep learning models (CNNs, LSTMs). Sensor fusion techniques will also be applied to effectively combine audio and motion data. The final phase of the project will involve a comparative analysis of all models (ML vs. DL, single-sensor vs. multi-sensor), the preparation of the final report, and a demonstration of the system.

## 6 Realistic Constraints

The success of this project depends on several factors:

- Availability of high-quality and well-labeled data from the four sensors.

- Noise in recordings, which may reduce classification accuracy [3].
- Synchronization of multimodal signals, which is technically challenging.
- Computational resources needed for audio spectrograms, accelerometer data, and strain signals [1].
- The risk of overfitting when working with limited datasets [4].
- Ethical requirements, including secure handling of sensitive audio data.

### **a. Social, Environmental, and Economic Impact**

This project can improve healthcare by providing objective cough and activity analysis, supporting early diagnosis and better treatment [2]. Socially, it can benefit patients by enabling remote monitoring and reducing unnecessary hospital visits. Environmentally, fewer hospital trips may help lower energy use and transportation emissions. Economically, the system could reduce costs for patients and healthcare providers by lowering the need for expensive diagnostic procedures [4].

### **b. Cost Analysis**

The only cost of the project is labor. Assuming the student researcher works about 10 hours per week during the semester at an hourly rate of \$15, the labor cost is estimated at around \$2400 per semester and about \$4800 for two semesters. Since the device is already available in the lab, no additional hardware expenses are expected.

### **c. Standards**

The system must follow ISO/IEEE 11073 standards for medical device communication and interoperability. On the software side, the development process should align with IEEE 12207 standards for software life cycle management. In addition, compliance with Turkish and EU Medical Device Regulations (MDR) is essential to ensure safety, privacy, and ethical use in healthcare.

## **7 Time Schedule**

The following monthly plan is tentative and may be adjusted according to project progress and data availability.

- **Month 1:** Literature review, background study, familiarization with device and dataset.

- **Month 2:** Data acquisition (multi-sensor), preprocessing (filtering, normalization, synchronization).
- **Month 3:** Segmentation (sliding windows), feature extraction (MFCCs, statistical features).
- **Month 4–5:** Baseline ML model creation and evaluation (SVM, RF, GB), cross-validation.
- **Month 6–7:** Deep learning model creation and evaluation (CNN, LSTM), sensor fusion, hyperparameter tuning.
- **Month 8:** Comparison of models (ML vs DL), robustness testing, final report preparation and real-time prototype demo (if time permits).

## References

- [1] J. Amoh and K. Odame, “Deep neural networks for identifying cough sounds,” *IEEE Trans. Biomed. Circuits Syst.*, vol. 10, no. 5, pp. 1003–1011, 2016.
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- [3] T. Otoshi, T. Nagano, S. Izumi, D. Hazama, *et al.*, “A novel automatic cough frequency monitoring system combining a triaxial accelerometer and a stretchable strain sensor,” *Sci. Rep.*, vol. 11, Art. no. 9973, 2021.
- [4] D. Sanchez-Morillo, D. Sales-Lerida, B. Priego-Torres, and A. León-Jiménez, “Cough detection using acceleration signals and deep learning techniques,” *Electronics*, vol. 13, no. 12, Art. no. 2410, 2024.